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CLAUDE CODE ANALYTICS PLATFORM

Project Overview

This analytics platform processes telemetry logs to provide actionable insights into API requests and tool usage patterns across your organisation.

Data Ingestion

Automated collection and cleaning of raw telemetry data from multiple sources

Analysis Engine

Comprehensive analysis of API requests, token usage, and tool performance metrics

Visualisation

Interactive dashboards and visualisations to track trends and key performance indicators

Folder Structure

Python scripts for data processing, analysis, and dashboard creation

src

1

2

data

Raw and processed telemetry data storage

CSV files and visualisation outputs

output

3

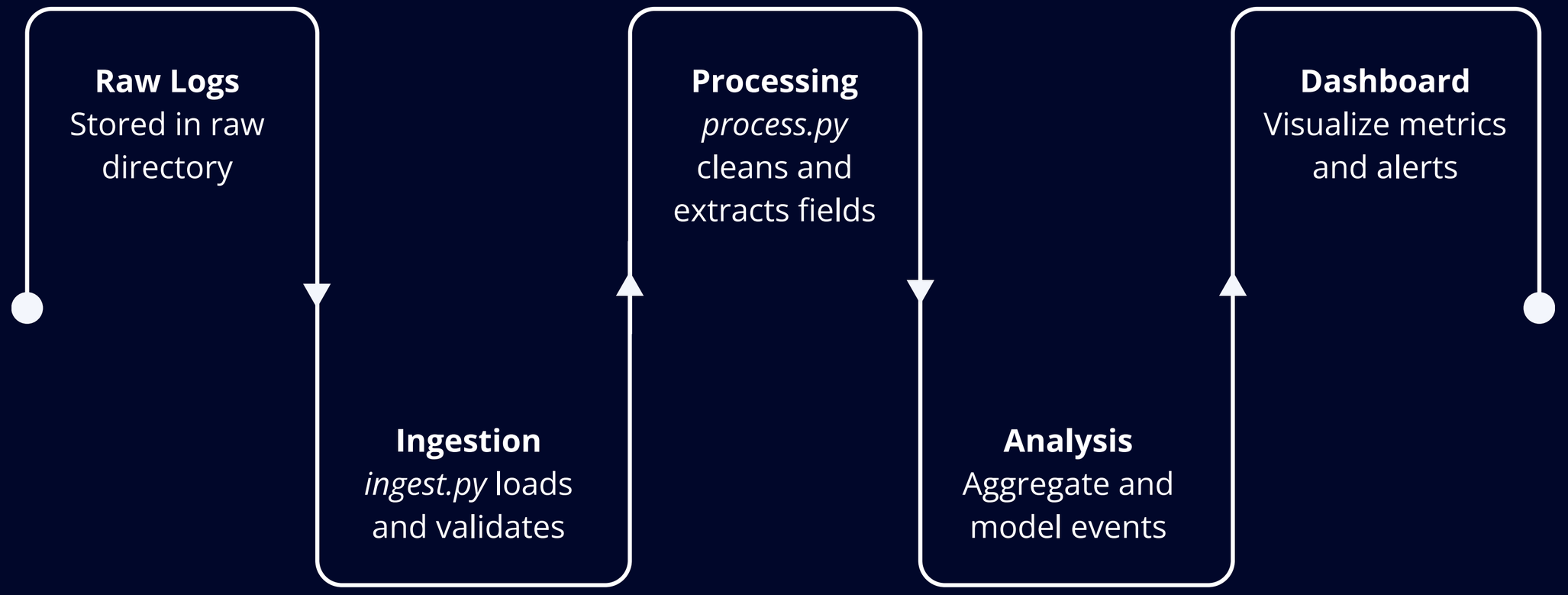
4

dashboard

Interactive dashboard application

Entry point: *main.py* orchestrates the complete analysis pipeline

Data Pipeline Architecture



Generated Outputs

llm_usage_summary.csv
Summarises token usage and associated costs across all API requests

predicted_usage.csv
Predicted usage data for capacity planning and resource allocation

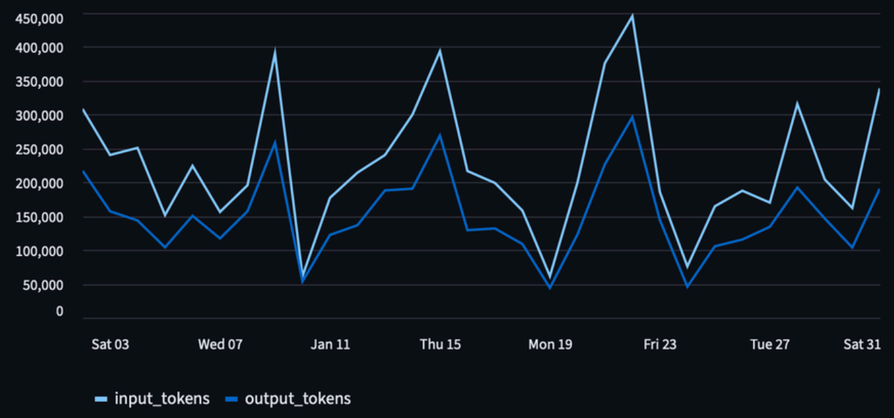
Visualisations

Framework
Built using Flask for responsive web interface

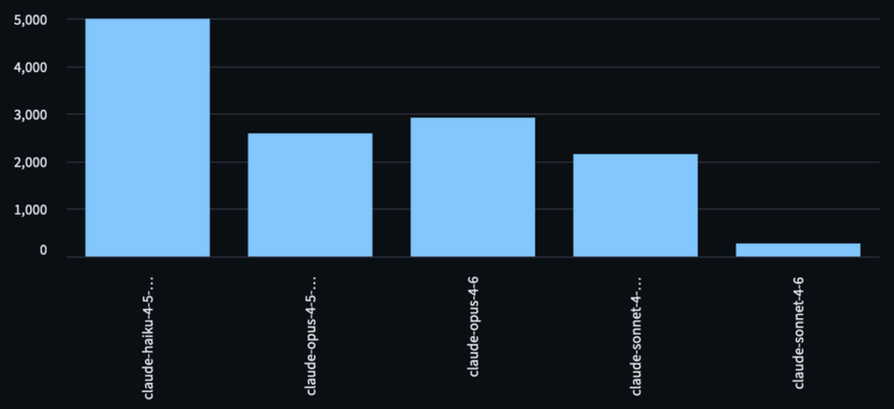
Entry Point
app.py launches the dashboard application

Launch Command
Run python *dashboard/app.py* to start server

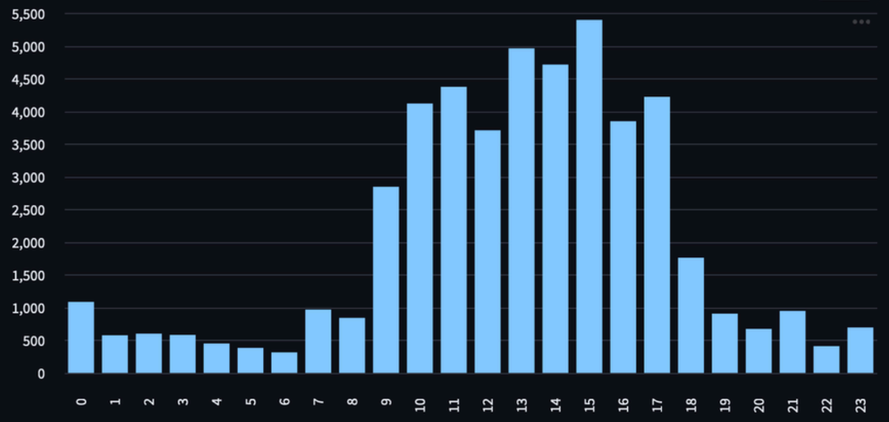
Token Usage Over Time



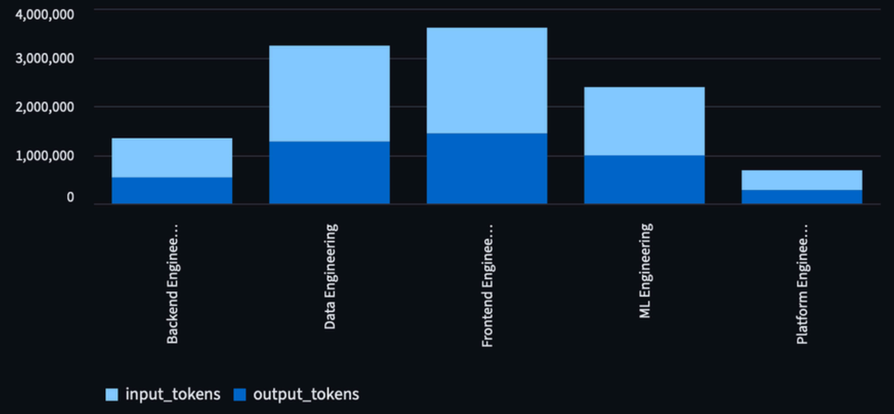
Top Models



Peak Usage Hours



Token Usage by Role



Requirements and Setup

1. **Python Version**
   3.8 or higher required for compatibility
2. **Core Libraries**
   pandas, matplotlib, seaborn, flask
3. **Virtual Environment**
   Create and activate isolated environment
4. **Install Dependencies**
   Run pip install -r requirements.txt

Key Prompts Used

Project Setup

- "Generate Python project structure with folders for ingestion, processing, analysis, and dashboard scripts."

Data Ingestion

- "Load telemetry JSON/CSV, handle missing values, and output cleaned DataFrame."

Data Processing

- "Normalize timestamps, remove duplicates, and aggregate token usage per user."

Analytics

- "Compute token consumption trends, peak usage times, and summarize LLM behavior in CSVs."

Dashboard

- "Create interactive dashboard showing usage trends and tool success metrics with Streamlit/Dash."

AI-Orchestration

- "Use LLMs to generate modular scripts, documentation, and automated outputs with minimal manual coding."

Thank You